

INTERNSHIP PRESENTATION

Grid-Forming & Grid-Following Inverter Control, and Dynamic Modeling of AI Data Center Load, in PSCAD

PCS average-model GFL/GFM control development — and a double-conversion UPS AI data center load model

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Road-Map

WORK STREAM A

GFL / GFM Control Module

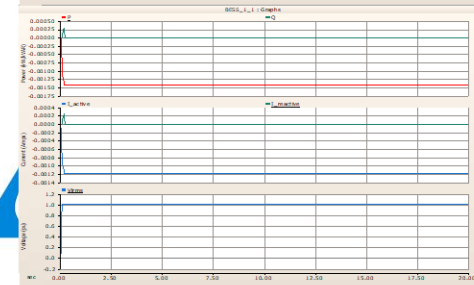
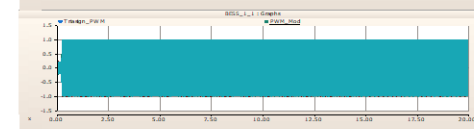
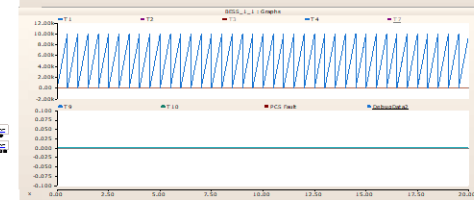
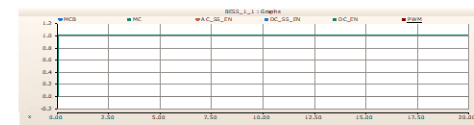
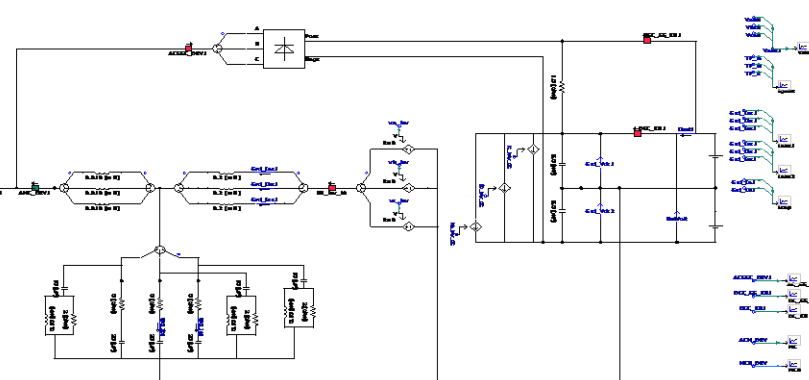
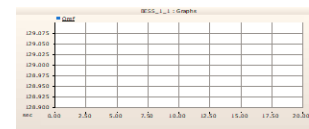
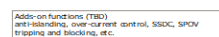
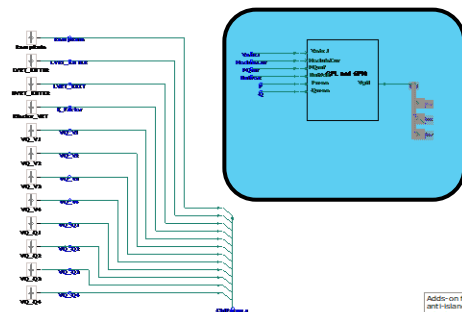
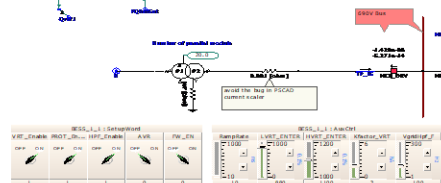
- 1 PCS - GFL/GFM Concept
- 2 SRF-PLL synchronization for GFL
- 3 Outer power loop, inner current loop, negative sequence
- 4 GFL results
- 5 GFM control: P-f and Q-V droop
- 6 GFM control: Virtual Synchronous Machine

WORK STREAM B

AI Data Center UPS Model

- 7 Motivation for AI Data Center Load modelling
- 8 Double-conversion UPS electrical topology
- 9 Rectifier (GFL) & inverter (GFM) control blocks
- 10 Bidirectional DC/DC converter & battery SOC
- 11 Voltage / frequency ride-through (VRT & FRT)

Figure 1: Schematic representation of the experimental design. The figure is divided into three main sections: 'Pre-Test', 'Main Experiment', and 'Post-Test'. Each section contains a timeline of tasks and a corresponding list of dependent variables. 'Pre-Test' includes a 10-min task and a 10-min rest, with dependent variables like 'Pre-Test Time' and 'Pre-Test Error'. 'Main Experiment' includes a 10-min task and a 10-min rest, with dependent variables like 'Main Experiment Time' and 'Main Experiment Error'. 'Post-Test' includes a 10-min task and a 10-min rest, with dependent variables like 'Post-Test Time' and 'Post-Test Error'. The tasks are labeled 'Task 1' and 'Task 2'.



PCS - GFL/GFM Concept

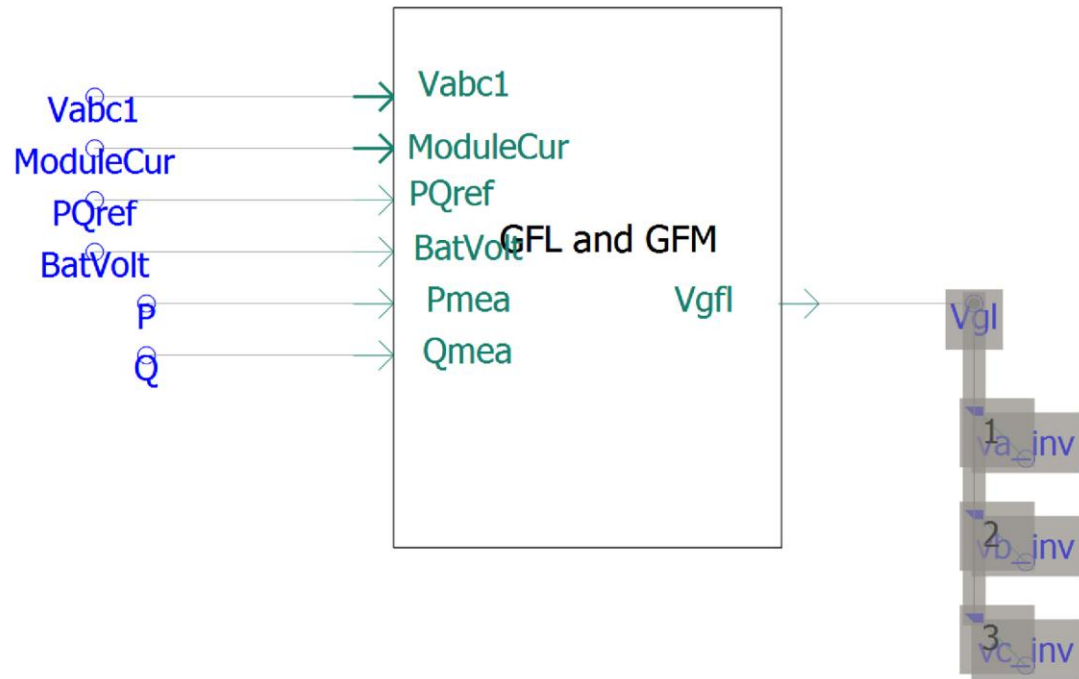


Figure: GFL/GFM module I/O inside the PCS average model

Grid-FOLLOWING (GFL)

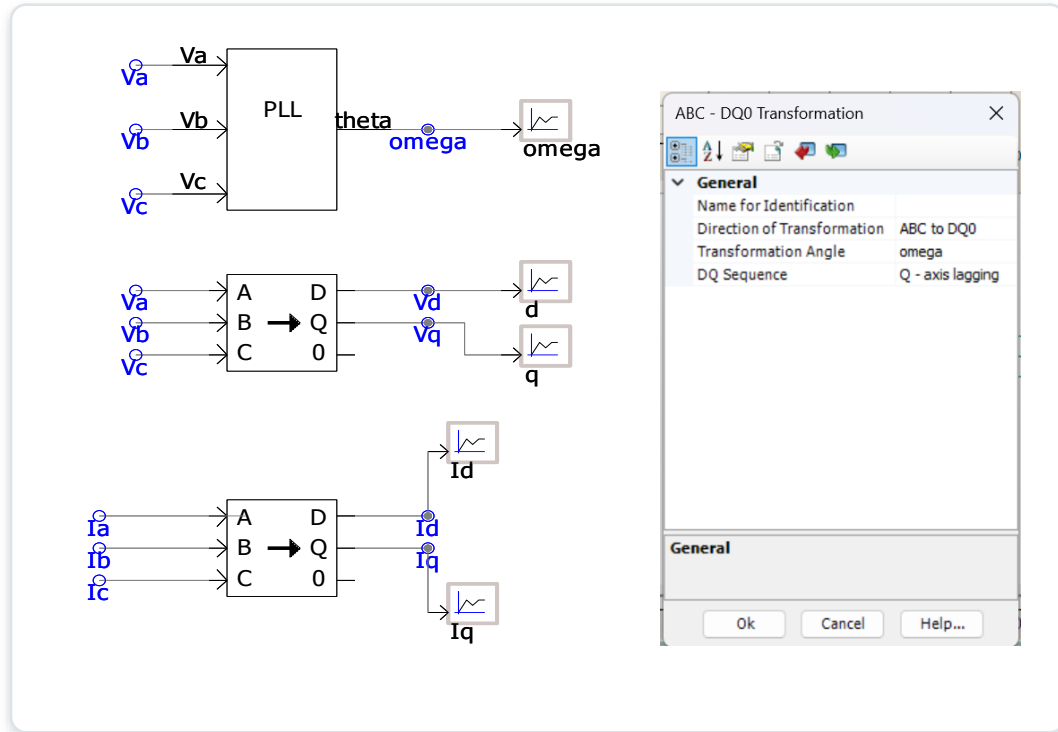
Controlled current source. Uses a PLL to track grid phase/frequency and injects current to P/Q references. Needs a strong external voltage to synchronize to.

Grid-FORMING (GFM)

Controlled voltage source with its own internal frequency/voltage reference — like a synchronous generator's rotor. Can black-start and support weak/islanded grids via droop or VSM control.



SRF-PLL Synchronization for GFL



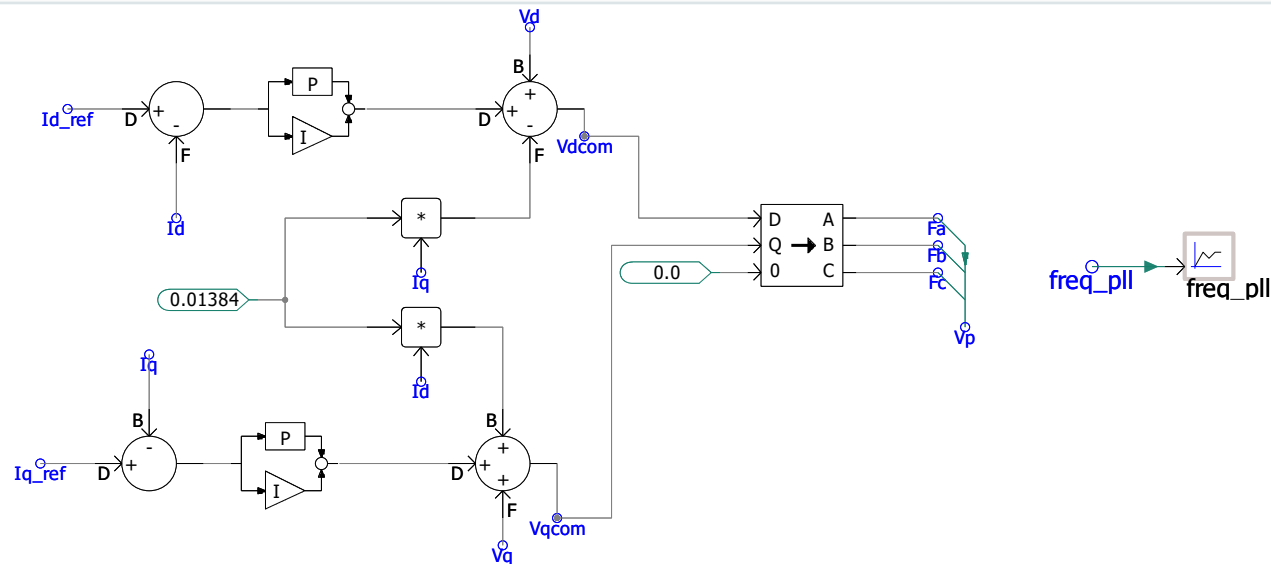
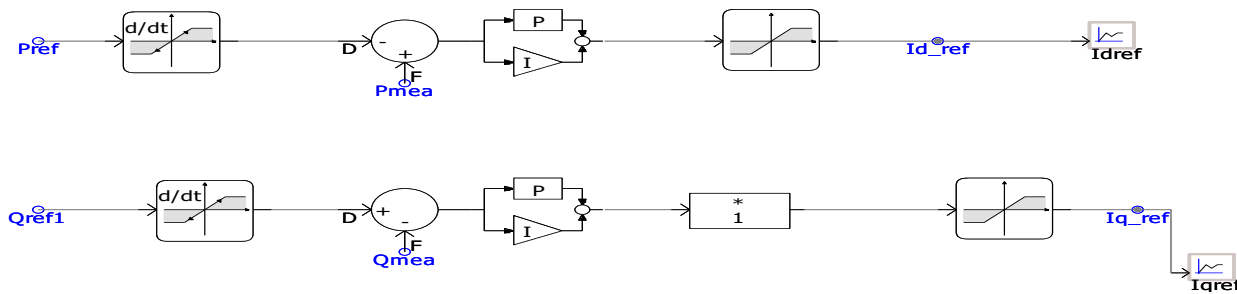
PLL bandwidth is tuned to grid strength (SCR)

Parameter	SCR 3	SCR 2	SCR 1.5
PLL BW	20 Hz	10 Hz	5 Hz
PLL Kp/Ki	177.7/15791	88.9/3948	44.4/987
Current Kp	0.121	0.184	0.246
P-loop Kp/Ki	0.15/60	0.10/35	0.06/20

As SCR drops 3 → 1.5, PLL bandwidth is deliberately cut 4× (20→5 Hz) to stay stable on a weaker grid — current-loop and power-loop gains are retuned together, not left fixed.

- Park/Clarke transform of V_a , V_b , V_c into the rotating dq frame
- PI controller drives the q-axis component to zero → locks the frame to the grid voltage vector
- Outputs θ (angle) and ω (frequency) that every other block is built around

Outer Power Loop & Inner Current Loop



- **Outer loop:**

- $P_{ref}/Q_{ref1} \rightarrow$ derivative/deadband $\rightarrow$ PI vs. $P_{mea}/Q_{mea} \rightarrow$ rate limiter $\rightarrow I_{dref} / I_{qref}$

- **Inner loop:**

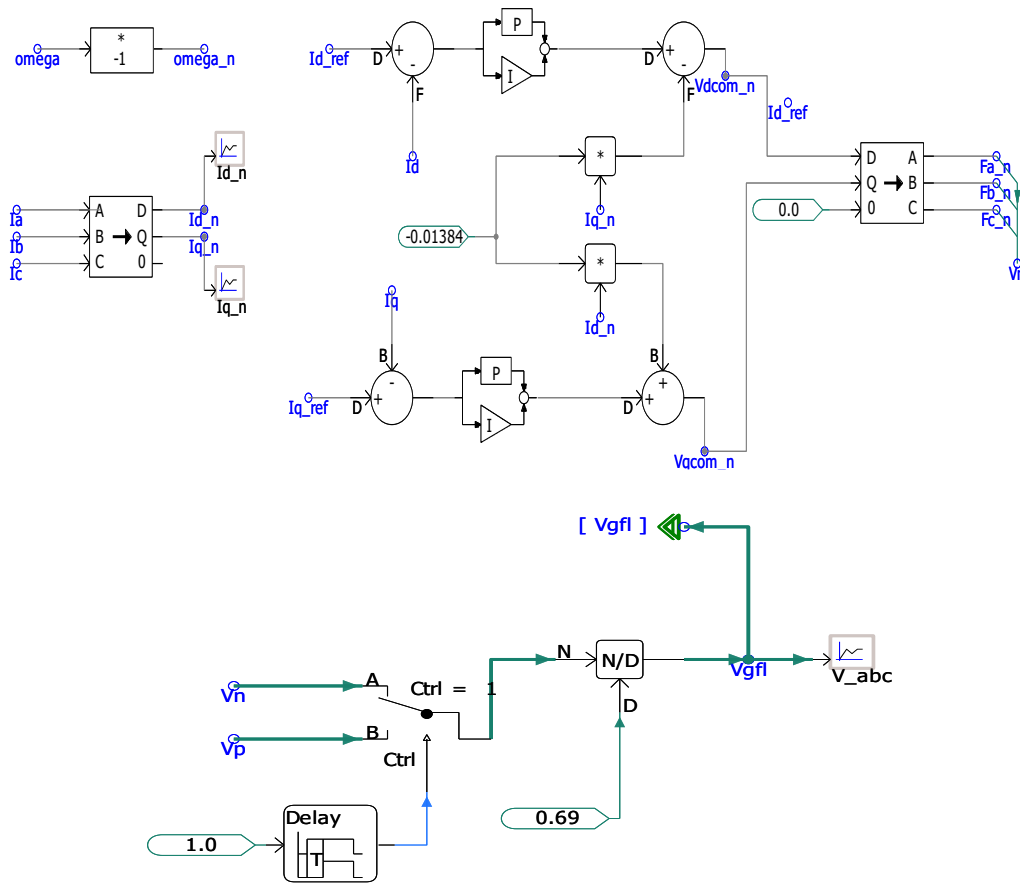
- abc $\rightarrow$ dq0 measurement, decoupled I_d/I_q PI control produces V_{dcom}, V_{qcom}

- **Decoupling:**

- 0.01384 pu feed-forward gains cancel d/q cross-coupling from the rotating frame



Negative-Sequence Control & Output Enable



Negative-sequence path

The entire dq current-control structure is mirrored using $\omega_n = -\omega$ and a separate I_{d_n}/I_{q_n} pair, so the converter can regulate negative-sequence current independently of positive-sequence current — needed for ride-through under unbalanced voltage sags rather than tripping.

Output-stage soft start

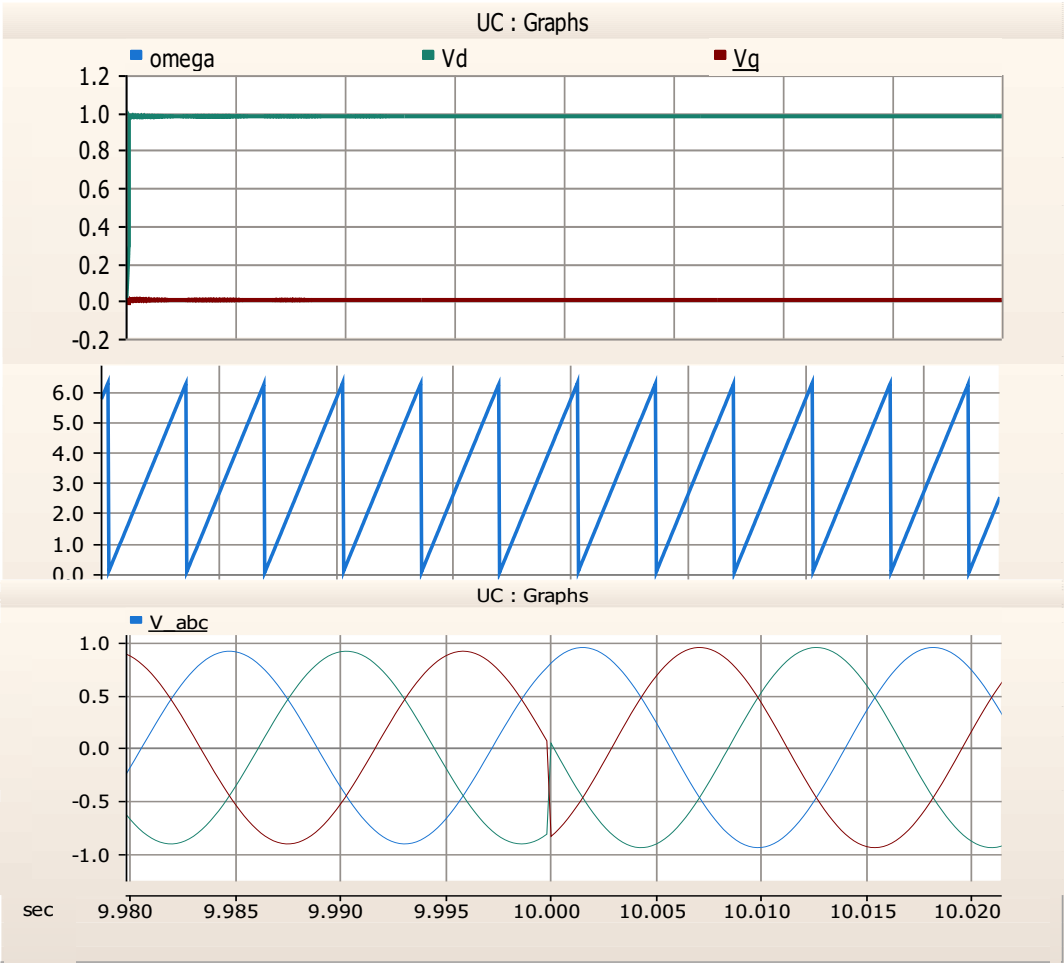
A Ctrl-driven switch, Delay block, and N/D divider assemble the dq voltage commands into the abc output V_{gfl} — ramping the output in cleanly on enable or mode change, rather than stepping.



GFL Results

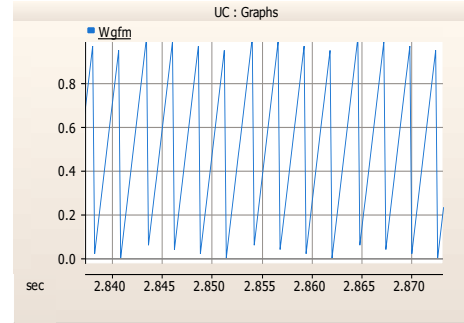
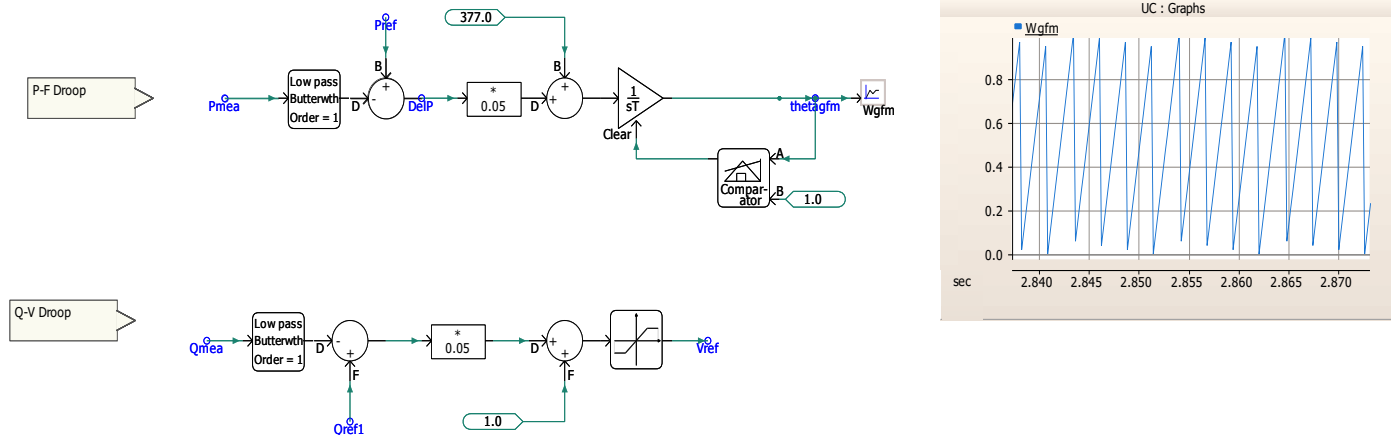


GFL results: Idref, Iqref, and Freq_PLL



GFL results: Omega, Vd, Vq, and V_abc

GFM Control: P-f and Q-V Droop



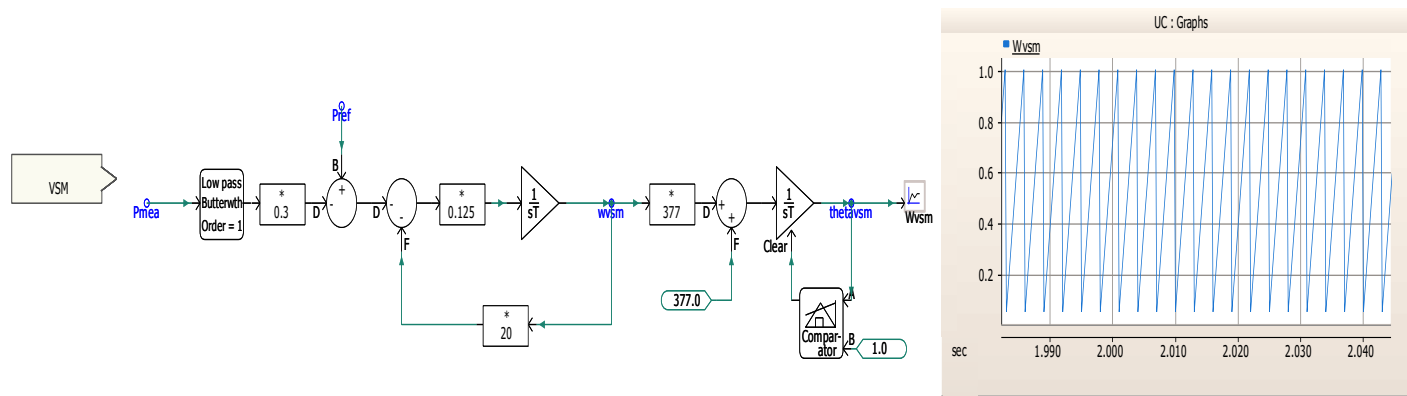
Reads like a generator

P-f droop mirrors a governor; Q-V droop mirrors an AVR. If $P_{mea} > P_{ref}$, frequency droops down in proportion to the 5% gain — exactly how parallel generators (or parallel GFM inverters) share load with no communication needed.

- **P-f droop:**
- P_{mea} (low-pass filtered) vs. P_{ref} , $\times 0.05$ droop gain (5%), added to 377 rad/s nominal, integrated with Comparator/Clear angle wraparound $\rightarrow$ θ_{tagfm}
- **Q-V droop:**
- Same structure on Q_{mea} vs. $Q_{ref1} \rightarrow V_{ref}$ (1.0 pu nominal, 5% gain, limiter)



GFM Control: Virtual Synchronous Machine



$$2H \cdot d\omega/dt = P_{ref} - P_{mea} - D \cdot (\omega - \omega_0)$$

- 0.125 gain $\leftrightarrow 1/(2H) \rightarrow$ inertia constant $H \approx 4$ s
- $\times 20$ feedback path $\leftrightarrow$ damping term $D = 20$
- Same 377 rad/s scaling + Comparator/Clear wraparound as the droop path $\rightarrow \theta_{vsm}$

Droop + inertia, not just droop

Because the swing equation carries an inertia term ($2H$), a sudden power imbalance changes VSM frequency gradually — a bounded rate-of-change-of-frequency (RoCoF) — rather than jumping toward the droop-implied point, while D still pulls it back to a comparable steady state.

WORK STREAM B

Dynamic Modeling of AI Data Center Load

A double-conversion UPS model



Why AI Data Center Load Modeling Matters

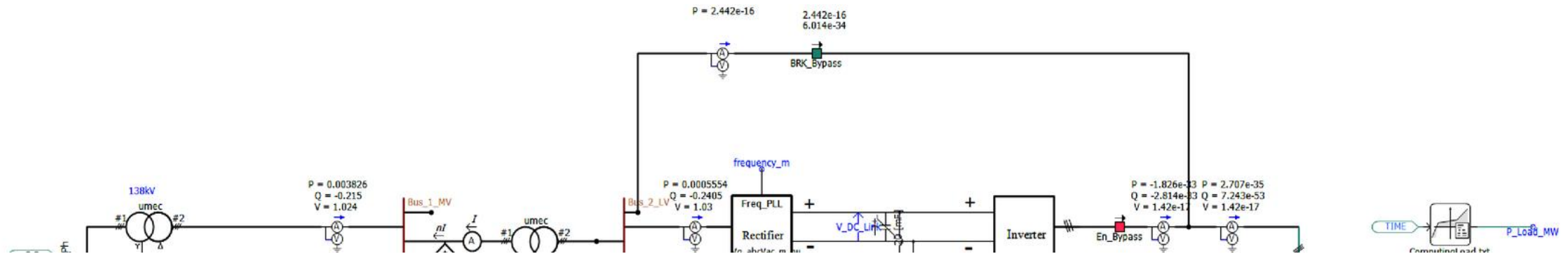
- AI data centers connect through UPS systems (rectifier, inverter, battery) with fast, nonlinear ride-through logic
- Documented events need to **(developing)** show that logic can disconnect a facility — or a fleet of them at once — on a fault that wouldn't trip a conventional load
- Formalized large-load interconnection is moving toward requiring detailed PSCAD/EMT converter models, not generic composite loads.

PRIMARY WORK, WORK STREAM B

Dynamic Modeling of AI Data Center Load in PSCAD

Open-source, modular PSCAD EMT model (switching + average) of a double-conversion UPS AI data center: rectifier, inverter, bidirectional DC/DC, battery, VFD cooling, computing loads, and a centralized VRT/FRT module. Published so industry has a shared, non-proprietary starting point.

Double-Conversion UPS Electrical Topology



Bypass path

Load connects directly to the utility bus through BRK_Bypass — no power-electronic conversion, most efficient, used when grid voltage/frequency are healthy.

Conversion path

Rectifier (AC→DC) → DC link → Inverter (DC→AC) continuously regulates 480V/60Hz to the load, isolating it from most upstream disturbances. Battery sits on the DC link via a bidirectional DC/DC converter.

Core Conversion Blocks: Rectifier & Inverter

RECTIFIER (AC → DC)

Grid-following converter

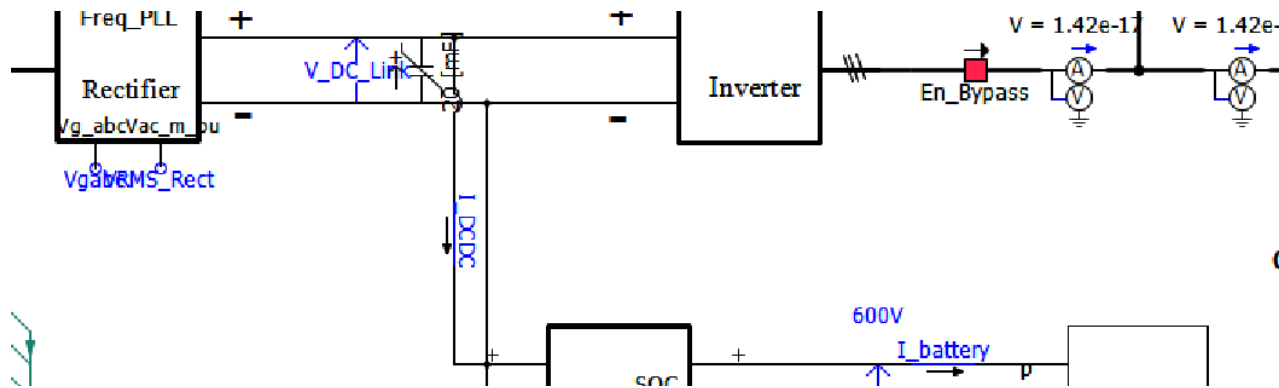
- 3-phase, 2-level VSC; synchronizes via PLL (Freq_PLL)
- dq double-loop cascade, same as Work Stream A
 - **d-axis:**
 - DC-link voltage (or active power during reconnection)
 - **q-axis:**
 - reactive power, typically $Q = 0$
- Ramp-rate limited on take-over, trip, and reconnection

INVERTER (DC → AC)

Grid-forming converter

- 3-phase, 2-level VSC; P-f / Q-V droop, same as Work Stream A
 - **d-axis:**
 - AC voltage magnitude
 - **q-axis:**
 - reference held at 0
- Forms its own voltage/frequency — the load only “sees” a disturbance if the DC link itself collapses
- Adds a VDC–VAC droop: reduces AC voltage as DC link sags, protecting the DC link

Bidirectional DC/DC Converter & Battery SOC



- **Battery / SOC block:**
- Non-ideal (Thevenin-like) model — internal source + output resistance, so terminal voltage sags under discharge
- SOC signal tracks remaining energy through charge/discharge cycling; essential for ride-through duration and future extensions (e.g., de-rating near SOC limits)

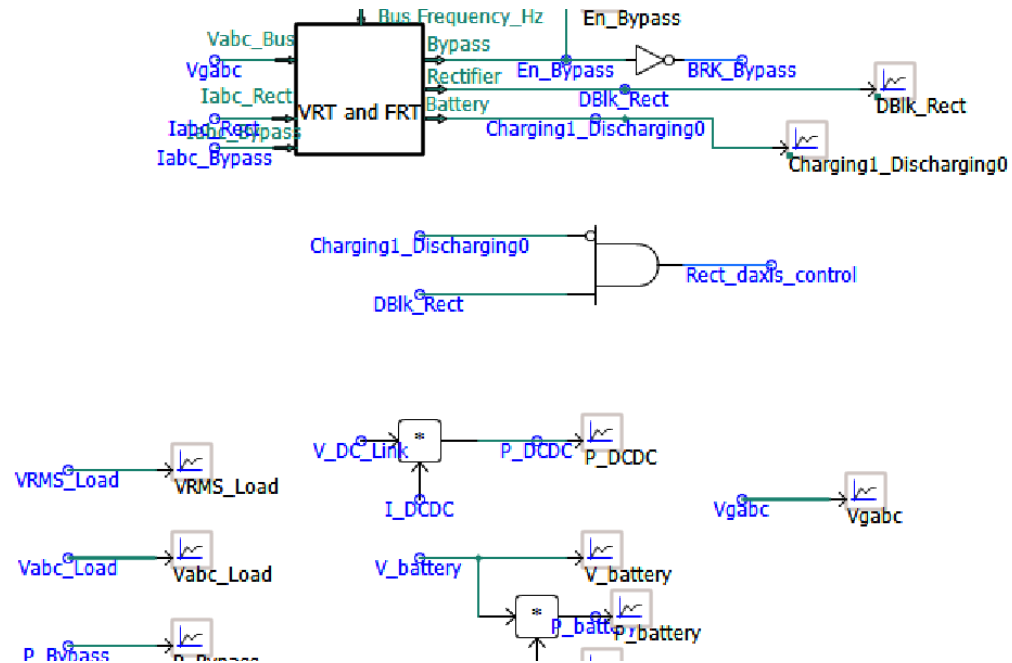
BUCK — charging

DC link → battery. Single-loop power control tracks a modest charging-power reference, well below the rectifier's spare capacity.

BOOST — discharging

Battery → DC link. Single-loop DC-link-voltage control — the battery becomes the DC link's primary regulator once the rectifier can't hold it up.

Voltage & Frequency Ride-Through (VRT/FRT)



The UPS's supervisory brain

- Monitors PCC voltage/frequency, rectifier current, bypass current against multi-point ride-through curves with hold-up times
- Drives 3 coordinated outputs: Bypass, DBlk_Rect, Charging1_Discharging0
- Rect_daxis_control AND-gate + first-order P_Inv_Setpoint filter smooth mode transitions
- Inverter is never blocked — the load never loses supply while sources negotiate

Three Operating Modes

Mode	Path to load	Rectifier	Battery
Bypass	Direct from utility, through BRK_Bypass	Idle / trickle-charging only	Standby or charging
Normal converter	Rectifier → DC link → Inverter	Regulates DC-link voltage (GFL)	Buck mode, charging
Battery backup	Battery → DC/DC → DC link → Inverter	Blocked / tripped	Boost mode, discharging, regulates DC link

Healthy grid → bypass (most efficient). Excursion beyond the bypass ride-through curve → normal converter mode (GFL rectifier + GFM inverter jointly regulate). Rectifier trip on a severe disturbance → battery backup keeps the DC link — and the inverter, and therefore the load — alive until the grid recovers and the rectifier reconnects on a load-like ramp.



Summary of Work Completed

WORK STREAM A

- GFL/GFM module integrated into the PCS average model with a defined I/O interface
- SRF-PLL synchronization tuned across 3 SCR operating points
- Outer power loop + inner current loop, with decoupling and full negative-sequence control
- GFM implemented two ways — P-f/Q-V droop and VSM — with reasoned RoCoF/inertia comparison

WORK STREAM B

- Double-conversion UPS topology reproduced from the ERCOT/TAMU reference model
- Rectifier mapped to GFL control, inverter to GFM droop — direct reuse of Stream A
- Bidirectional DC/DC buck/boost logic and battery/SOC block documented
- Centralized VRT/FRT module and 3 operating modes documented end to end

Key Learnings & Next Steps

LEARNINGS

- PLL bandwidth is a stability lever co-tuned with grid strength, not a fixed constant
- GFM's real value is giving downstream loads/buses something to synchronize to
- VSM = droop + explicit inertia — bounded RoCoF during the transient
- UPS ride-through is a coordinated, multi-converter behavior, not one block's job

NEXT STEPS

- Run the PSCAD cases; replace every illustrative chart with captured data
- Validate PLL/current-loop tuning with small-signal stability margins
- Multi-disturbance VSM vs. droop study — quantify RoCoF & nadir improvement
- Exercise the full LVRT/HVRT curve; load the real ComputingLoad.txt profile



Thank You

Questions & Discussion

